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Academic Proffesionalism

ARS Draft Period

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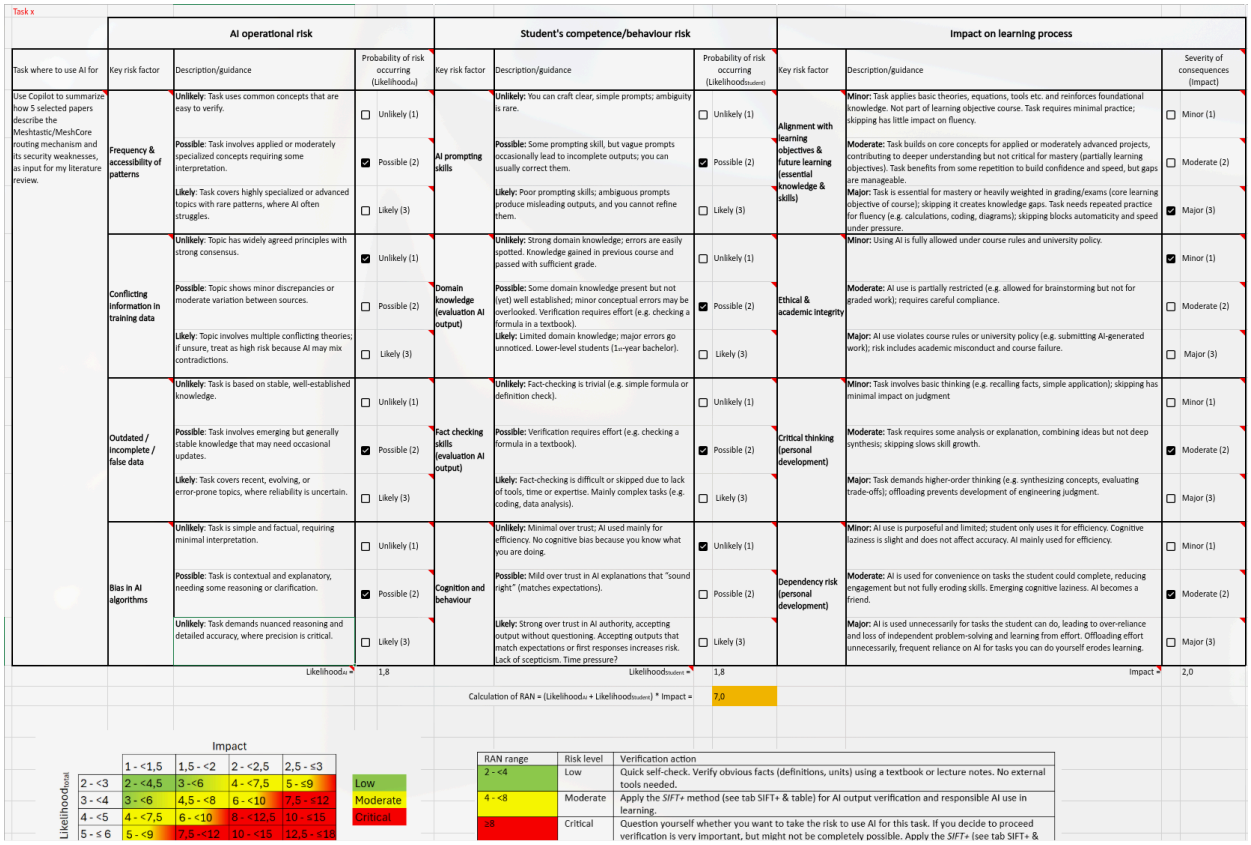
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1. AI usage and risk assessment

I used generative AI for several tasks in my parts of the literature review (Introduction, Sections 2.1 and 2.3, Appendix A) and of the research proposal. Before using it, I assessed a representative task with the AI Risk Assessment Protocol (AI-RAP): *using AI to summarise how five selected papers describe the Meshtastic/MeshCore routing mechanism and its security weaknesses, as input for my literature review.* Figure 1 shows my completed worksheet.



Task x

	AI operational risk			Student's competence/behaviour risk			Impact on learning process		
Task where to use AI for	Key risk factor	Description/guidance	Probability of risk occurring (Likelihood$_{AI}$)	Key risk factor	Description/guidance	Probability of risk occurring (Likelihood$_{student}$)	Key risk factor	Description/guidance	Severity of consequences (Impact)
Use Copilot to summarize how 5 selected papers describe the Meshtastic/MeshCore routing mechanism and its security weaknesses, as input for my literature review.	Frequency & accessibility of patterns	**Unlikely**: Task uses common concepts that are easy to verify.	☐ Unlikely (1)	AI prompting skills	**Unlikely:** You can craft clear, simple prompts; ambiguity is rare.	☐ Unlikely (1)	Alignment with learning objectives & future learning (essential knowledge & skills)	**Minor:** Task applies basic theories, equations, tools etc. and reinforces foundational knowledge. Not part of learning objective course. Task requires minimal practice; skipping has little impact on fluency.	☐ Minor (1)
		Possible: Task involves applied or moderately specialized concepts requiring some interpretation.	☑ Possible (2)		**Possible:** Some prompting skill, but vague prompts occasionally lead to incomplete outputs; you can usually correct them.	☑ Possible (2)		**Moderate:** Task builds on core concepts for applied or moderately advanced projects, contributing to deeper understanding but not critical for mastery (partially learning objectives). Task benefits from some repetition to build confidence and speed, but gaps are manageable.	☐ Moderate (2)
		Likely: Task covers highly specialized or advanced topics with rare patterns, where AI often struggles.	☐ Likely (3)		**Likely:** Poor prompting skills; ambiguous prompts produce misleading outputs, and you cannot refine them.	☐ Likely (3)		**Major:** Task is essential for mastery or heavily weighted in grading/exams (core learning objective of course); skipping it creates knowledge gaps. Task needs repeated practice for fluency (e.g. calculations, coding, diagrams); skipping blocks automaticity and speed under pressure.	☑ Major (3)
	Conflicting information in training data	**Unlikely**: Topic has widely agreed principles with strong consensus.	☑ Unlikely (1)	Domain knowledge (evaluation AI output)	**Unlikely:** Strong domain knowledge; errors are easily spotted. Knowledge gained in previous course and passed with sufficient grade.	☐ Unlikely (1)	Ethical & academic integrity	**Minor:** Using AI is fully allowed under course rules and university policy.	☑ Minor (1)
		Possible: Topic shows minor discrepancies or moderate variation between sources.	☐ Possible (2)		**Possible:** Some domain knowledge present but not (yet) well established; minor conceptual errors may be overlooked. Verification requires effort (e.g. checking a formula in a textbook).	☑ Possible (2)		**Moderate:** AI use is partially restricted (e.g. allowed for brainstorming but not for graded work); requires careful compliance.	☐ Moderate (2)
		Likely: Topic involves multiple conflicting theories; if unsure, treat as high risk because AI may mix contradictions.	☐ Likely (3)		**Likely:** Limited domain knowledge; major errors go unnoticed. Lower-level students (1st-year bachelor).	☐ Likely (3)		**Major:** AI use violates course rules or university policy (e.g. submitting AI-generated work); risk includes academic misconduct and course failure.	☐ Major (3)
	Outdated / incomplete / false data	**Unlikely**: Task is based on stable, well-established knowledge.	☐ Unlikely (1)	Fact checking skills (evaluation AI output)	**Unlikely:** Fact-checking is trivial (e.g. simple formula or definition check).	☐ Unlikely (1)	Critical thinking (personal development)	**Minor:** Task involves basic thinking (e.g. recalling facts, simple application); skipping has minimal impact on judgment	☐ Minor (1)
		Possible: Task involves emerging but generally stable knowledge that may need occasional updates.	☑ Possible (2)		**Possible:** Verification requires effort (e.g. checking a formula in a textbook).	☑ Possible (2)		**Moderate:** Task requires some analysis or explanation, combining ideas but not deep synthesis; skipping slows skill growth.	☑ Moderate (2)
		Likely: Task covers recent, evolving, or error-prone topics, where reliability is uncertain.	☐ Likely (3)		**Likely:** Fact-checking is difficult or skipped due to lack of tools, time or expertise. Mainly complex tasks (e.g. coding, data analysis).	☐ Likely (3)		**Major:** Task demands higher-order thinking (e.g. synthesizing concepts, evaluating trade-offs); offloading prevents development of engineering judgment.	☐ Major (3)
	Bias in AI algorithms	**Unlikely**: Task is simple and factual, requiring minimal interpretation.	☐ Unlikely (1)	Cognition and behaviour	**Unlikely:** Minimal over trust; AI used mainly for efficiency. No cognitive bias because you know what you are doing.	☑ Unlikely (1)	Dependency risk (personal development)	**Minor:** AI use is purposeful and limited; student only uses it for efficiency. Cognitive laziness is slight and does not affect accuracy. AI mainly used for efficiency.	☐ Minor (1)
		Possible: Task is contextual and explanatory, needing some reasoning or clarification.	☑ Possible (2)		**Possible:** Mild over trust in AI explanations that "sound right" (matches expectations).	☐ Possible (2)		**Moderate:** AI is used for convenience on tasks the student could complete, reducing engagement but not fully eroding skills. Emerging cognitive laziness. AI becomes a friend.	☑ Moderate (2)
		Unlikely: Task demands nuanced reasoning and detailed accuracy, where precision is critical.	☐ Likely (3)		**Likely:** Strong over trust in AI authority, accepting output without questioning. Accepting outputs that match expectations or first responses increases risk. Lack of scepticism. Time pressure?	☐ Likely (3)		**Major:** AI is used unnecessarily for tasks the student can do, leading to over-reliance and loss of independent problem-solving and learning from effort. Offloading effort unnecessarily, frequent reliance on AI for tasks you can do yourself erodes learning.	☐ Major (3)
		Likelihood$_{AI}$ =	1,8		Likelihood$_{student}$ =	1,8		Impact =	2,0
					Calculation of RAN = (Likelihood$_{AI}$ + Likelihood$_{student}$) * Impact =	7,0			

Likelihood$_{total}$ \ Impact	1 - <1,5	1,5 - <2	2 - <2,5	2,5 - ≤3
2 - <3	2 - <4,5	3 - <6	4 - <7,5	5 - ≤9
3 - <4	3 - <6	4,5 - <8	6 - <10	7,5 - ≤12
4 - <5	4 - <7,5	6 - <10	8 - <12,5	10 - ≤15
5 - ≤ 6	5 - <9	7,5 - <12	10 - <15	12,5 - ≤18

Low
Moderate
Critical

RAN range	Risk level	Verification action
2 - <4	Low	Quick self-check. Verify obvious facts (definitions, units) using a textbook or lecture notes. No external tools needed.
4 - <8	Moderate	Apply the *SIFT+* method (see tab SIFT+ & table) for AI output verification and responsible AI use in learning.
≥8	Critical	Question yourself whether you want to take the risk to use AI for this task. If you decide to proceed verification is very important, but might not be completely possible. Apply the *SIFT+* (see tab SIFT+ &

Figure 1: My completed AI-RAP worksheet for summarising papers with AI.

The scores I gave per key risk factor (1 = low, 3 = high) are:

- **AI operational risk:** frequency and accessibility of patterns 2, conflicting information in training data 1, outdated, incomplete or false data 2, bias in AI algorithms 2.
- **My competence and behaviour:** AI prompting skills 2, domain knowledge (evaluating AI output) 2, fact-checking skills 2, cognition and behaviour 1.
- **Impact on learning process:** alignment with learning objectives 3, ethical and academic integrity 1, critical thinking 2, dependency risk 2.

This gives $L_{\mathrm{AI}} = 1.8$, $L_{\mathrm{S}} = 1.8$ and $I = 2.0$, so $\mathrm{RAN} = (1.8 + 1.8) \times 2.0 = 7.0$. That is a **moderate** risk, for which the protocol prescribes verification with the SIFT+ method. The highest single score is for alignment with the learning objectives: summarising and evaluating literature is exactly what this course is meant to teach me. Recent and specialised topics such as Meshtastic and MeshCore also make outdated or incomplete AI information likely.

Verification. Following SIFT+, I applied these checks to all my AI use:

- **Investigate the source:** every source came from my own Scopus searches or from official documentation and datasheets; no reference was taken from AI output alone.
- **Trace claims to the original context:** I checked summarised claims against the original PDFs and documentation pages.
- **Human checkpoint:** I reviewed every AI-suggested screening decision for the 509 records, and I ran and refined the search strings in Scopus myself. S3 returned many false positives ("LoRA" as Low-Rank Adaptation), so I refined it into S3b. The AI wrote the original query, but I was the one who spotted that the results were wrong. This showed me that I must always check what a query returns. Because so much research exists on AI, it was easy to see that these results were off-topic.

These checks caught real errors, such as a study attributed to the wrong authors and a truncated DOI.

Reflection. Besides summarising, I also used AI for first-pass screening and for drafting text. These tasks are even closer to the learning aims of reviewing literature and writing scientifically. I would therefore score them higher on impact, which would likely put them at a critical RAN. The main risk I noticed was not invented sources, but plausible-sounding text that slightly misrepresented a paper or named the wrong author, and I only found these errors by going back to the originals. I also noticed that I learned less from the writing itself than I would have otherwise. For the paper in Q2 I will write the text myself and use AI only for feedback on my own drafts and for checking code. This lowers the impact score and keeps the learning with me.

2. Meeting & supervision

Before contacting our supervisor, Geert Heijenk, my teammate and I first formulated a research question and sub-questions, so that the first meeting could focus on content. Geert replied by email with feedback on what our validation plan should look like, which we discussed further in the meeting.

His main feedback in the meeting, and what I did with it:

- **Our research question and sub-questions were too broad.** He expected that the question would become more specific after several iterations of reading and drafting. This turned out to be right: after reading the literature I realised the question was too broad, and we narrowed it from secure routing in general to the cost of hop-by-hop authenticated encryption on LoRa relay nodes across microcontroller classes. This narrower question fits my embedded-systems background, because it can be answered with measurements on real hardware. I will ask for feedback on it in the next meeting.
- **Research is iterative, not linear.** We had divided the work into fixed steps that we planned to complete one after another. He made clear that the scope, the literature review and the proposal develop together through many iterations.
- **Define a scope first, then iterate.** I was unsure at which point the scoping and the PRISMA process should start. He explained that the scope should be defined before

the literature review and proposal and then refined in each iteration. I applied this by fixing the scope (peer-to-peer LoRa mesh networks and comparable low-power multi-hop networks) and the inclusion and exclusion criteria before screening, and by adding two criteria (EC6 and EC7) during screening when it became necessary.

- **Expect to read a few dozen papers.** Because our topic is specialised, he told us that we would need at least a few dozen papers before the picture becomes consistent. My Sections 2.1 and 2.3 are based on 21 studies and 7 grey-literature sources, selected from 509 screened records.

On 12 October I sent Prof. Heijenk our research proposal and literature review draft, together with the course number he had asked for. At the time of writing we have not yet received a reply or feedback on these drafts. I will follow up with him before the next deadline, so that his feedback can still be used in the next iteration.

3. Teamwork

My teammate and I divided the work by research question. We have four sub-questions in total: two on cybersecurity and two on embedded systems. I worked on the two embedded-systems questions. We planned to review each other's sections, but because of time constraints from other work we have not done this yet. I will schedule it before the next deadline.

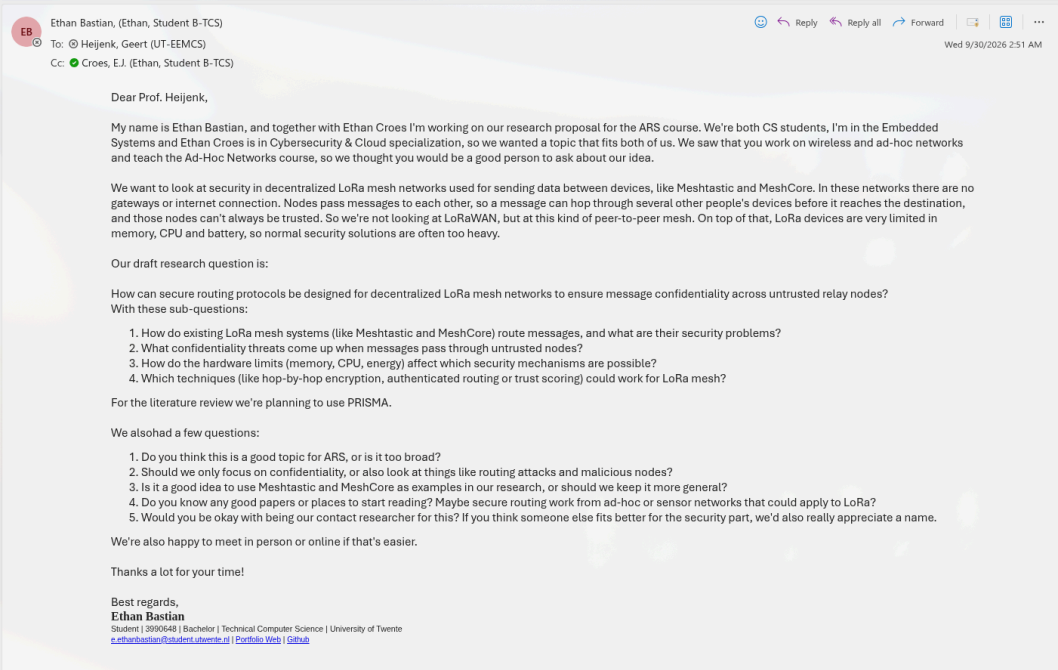
Besides working with my teammate, I peer-reviewed the literature reviews and research proposals of two other students, Levi and Luiz, on 6 November. I sent each of them the filled-in review form together with a short summary of their strongest points and the main things to improve. For Levi I could only review the literature review, because I had not received his research proposal.

4. Reflection

I wrote this assignment in a short time, next to my other work, Capita Selecta and a maths course that I find challenging. My time management was not good enough, and this made me depend more on AI than I wanted. Spotting the wrong S3 results and the wrong author showed me that AI output needs checking against the original source, and that this checking takes time I should plan for. In the next quartile I will start earlier, write the text myself and use AI only for feedback on my own drafts. I will also have my teammate review my sections, and I will keep notes of meetings and decisions as I go instead of collecting them at the end.

5. Evidence

The emails below show how I contacted my supervisor and how the supervision was arranged, in chronological order. We found the topic through the ARS assignment CS9. Because the coordinator had named Suzan Bayhan as our contact, I informed her too, and she confirmed that Prof. Heijenk could supervise us.



Ethan Bastian, (Ethan, Student B-TCS)
To: Heijenk, Geert (UT-EEMCS)
Cc: Croes, E.J. (Ethan, Student B-TCS)

Reply Reply all Forward
Wed 9/30/2026 2:51 AM

Dear Prof. Heijenk,

My name is Ethan Bastian, and together with Ethan Croes I'm working on our research proposal for the ARS course. We're both CS students, I'm in the Embedded Systems and Ethan Croes is in Cybersecurity & Cloud specialization, so we wanted a topic that fits both of us. We saw that you work on wireless and ad-hoc networks and teach the Ad-Hoc Networks course, so we thought you would be a good person to ask about our idea.

We want to look at security in decentralized LoRa mesh networks used for sending data between devices, like Meshtastic and MeshCore. In these networks there are no gateways or internet connection. Nodes pass messages to each other, so a message can hop through several other people's devices before it reaches the destination, and those nodes can't always be trusted. So we're not looking at LoRaWAN, but at this kind of peer-to-peer mesh. On top of that, LoRa devices are very limited in memory, CPU and battery, so normal security solutions are often too heavy.

Our draft research question is:

How can secure routing protocols be designed for decentralized LoRa mesh networks to ensure message confidentiality across untrusted relay nodes?
With these sub-questions:

1. How do existing LoRa mesh systems (like Meshtastic and MeshCore) route messages, and what are their security problems?
2. What confidentiality threats come up when messages pass through untrusted nodes?
3. How do the hardware limits (memory, CPU, energy) affect which security mechanisms are possible?
4. Which techniques (like hop-by-hop encryption, authenticated routing or trust scoring) could work for LoRa mesh?

For the literature review we're planning to use PRISMA.

We alsohad a few questions:

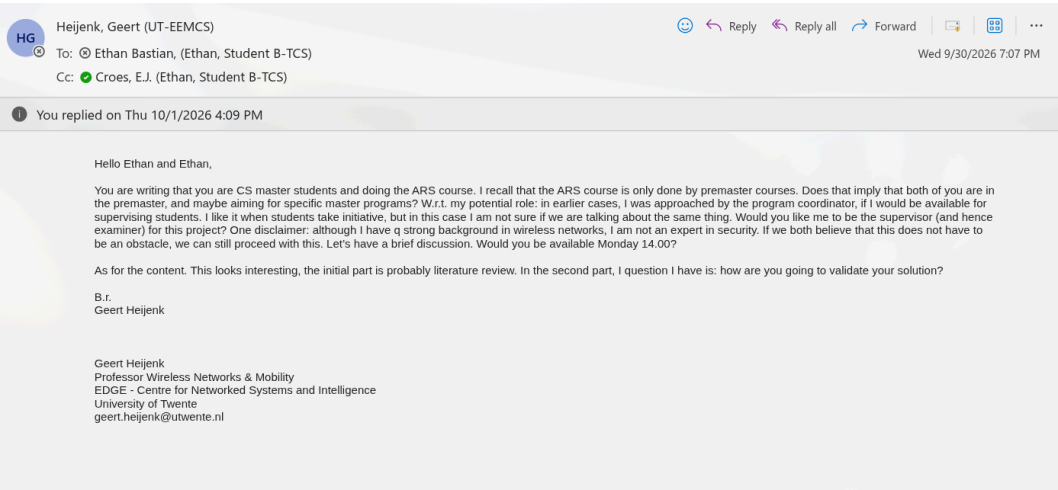
1. Do you think this is a good topic for ARS, or is it too broad?
2. Should we only focus on confidentiality, or also look at things like routing attacks and malicious nodes?
3. Is it a good idea to use Meshtastic and MeshCore as examples in our research, or should we keep it more general?
4. Do you know any good papers or places to start reading? Maybe secure routing work from ad-hoc or sensor networks that could apply to LoRa?
5. Would you be okay with being our contact researcher for this? If you think someone else fits better for the security part, we'd also really appreciate a name.

We're also happy to meet in person or online if that's easier.

Thanks a lot for your time!

Best regards,
Ethan Bastian
Student | 3990648 | Bachelor | Technical Computer Science | University of Twente
e.ethanbastian@student.utwente.nl | Portfolio Web | Github

Figure 2: 30 Sep: my first email to Prof. Heijenk, with our draft research question, sub-questions and five questions about the topic.



Heijenk, Geert (UT-EEMCS)
To: Ethan Bastian, (Ethan, Student B-TCS)
Cc: Croes, E.J. (Ethan, Student B-TCS)

Reply Reply all Forward
Wed 9/30/2026 7:07 PM

You replied on Thu 10/1/2026 4:09 PM

Hello Ethan and Ethan,

You are writing that you are CS master students and doing the ARS course. I recall that the ARS course is only done by premaster courses. Does that imply that both of you are in the premaster, and maybe aiming for specific master programs? W.r.t. my potential role: in earlier cases, I was approached by the program coordinator, if I would be available for supervising students. I like it when students take initiative, but in this case I am not sure if we are talking about the same thing. Would you like me to be the supervisor (and hence examiner) for this project? One disclaimer: although I have q strong background in wireless networks, I am not an expert in security. If we both believe that this does not have to be an obstacle, we can still proceed with this. Let's have a brief discussion. Would you be available Monday 14.00?

As for the content. This looks interesting, the initial part is probably literature review. In the second part, I question I have is: how are you going to validate your solution?

B.r.
Geert Heijenk

Geert Heijenk
Professor Wireless Networks & Mobility
EDGE - Centre for Networked Systems and Intelligence
University of Twente
geert.heijenk@utwente.nl

Figure 3: 30 Sep: his reply, asking about his role as supervisor and how we would validate our solution.

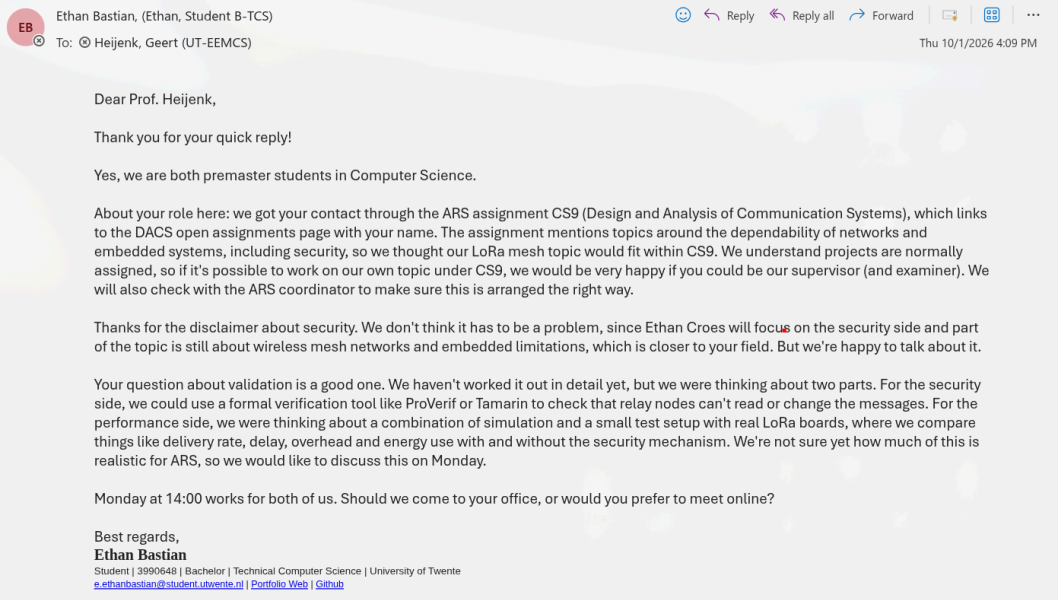


Figure 4: 1 Oct: my reply, with a first idea for validation (formal verification).

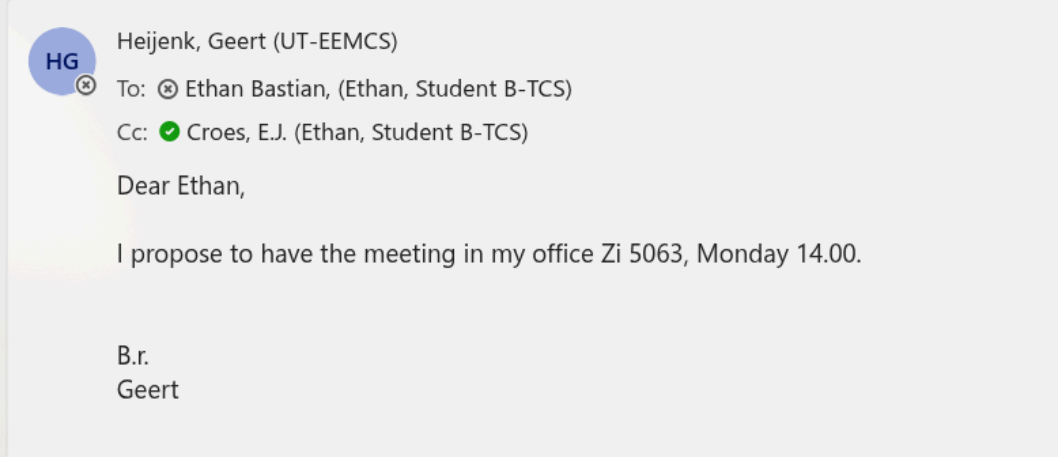


Figure 5: His proposal for the meeting in his office on Monday at 14:00.



Figure 6: Meeting request from Prof. Heijenk, which I accepted.

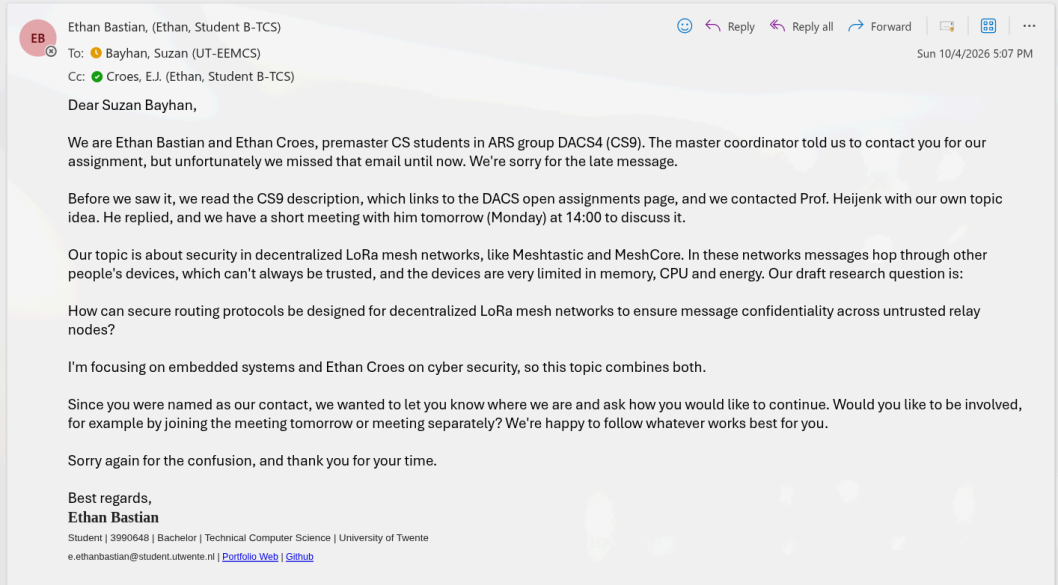


Figure 7: 4 Oct: my email to Suzan Bayhan, the contact named by the coordinator, explaining our situation and apologising for the late message.

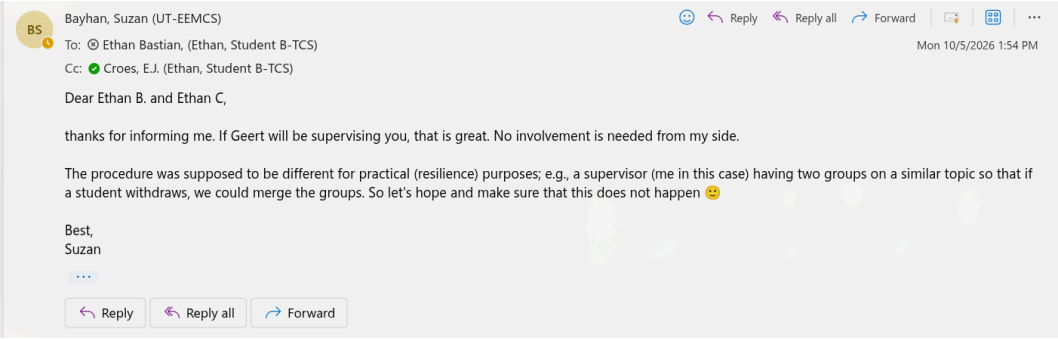


Figure 8: 5 Oct: her reply, confirming that no involvement is needed from her side.

The last three emails show what happened after the first meeting. On 12 October I sent Prof. Heijenk the course number he had asked for, together with our research proposal and literature review drafts, and apologised for sending it later than promised. We have not received a reply or any feedback on these drafts yet, so I will send a short follow-up and ask whether we can discuss the drafts in a next meeting. On 6 November I sent my peer reviews to Levi and Luiz. For each of them I filled in the review form and wrote a short summary in the email: first what was strong in their work, then the main points to improve. For Levi I could only review the literature review, because I had not received his research proposal, and I told him this openly. Writing these reviews also helped me look more critically at our own drafts, for example at whether our screening numbers and our gap are as clear as I asked from them.

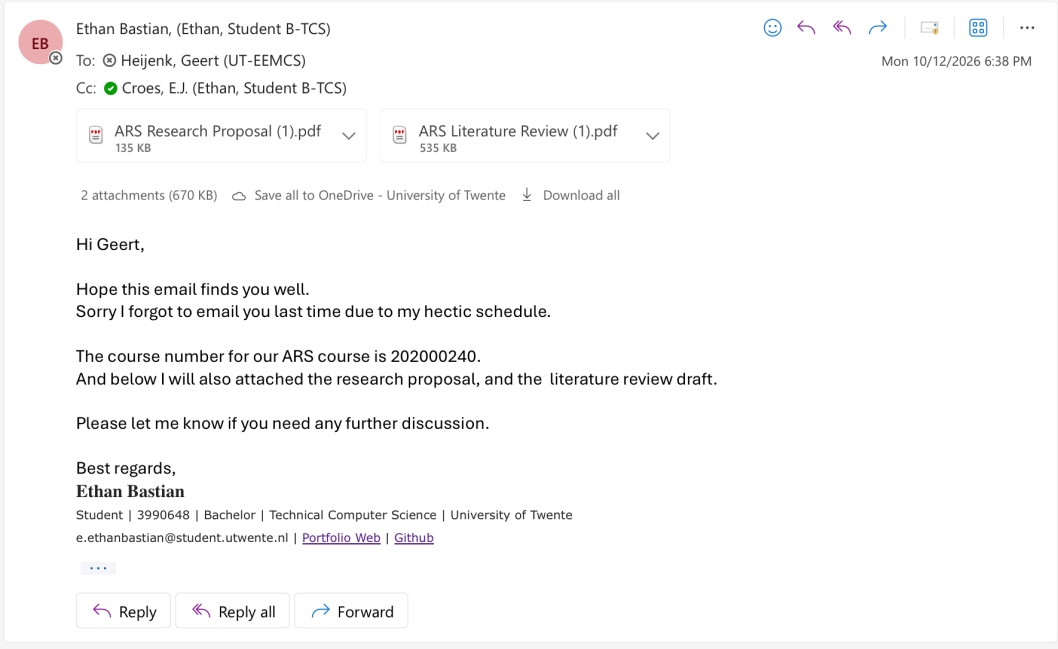


Figure 9: 12 Oct: my email to Prof. Heijenk with the course number and our research proposal and literature review drafts. No reply has been received yet.

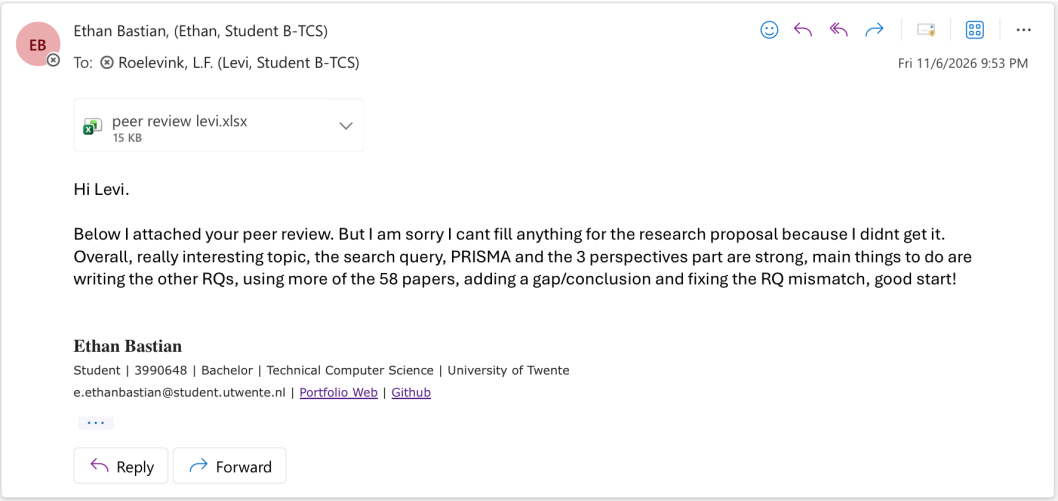


Figure 10: 6 Nov: my peer review sent to Levi, with a summary of strengths and points to improve.

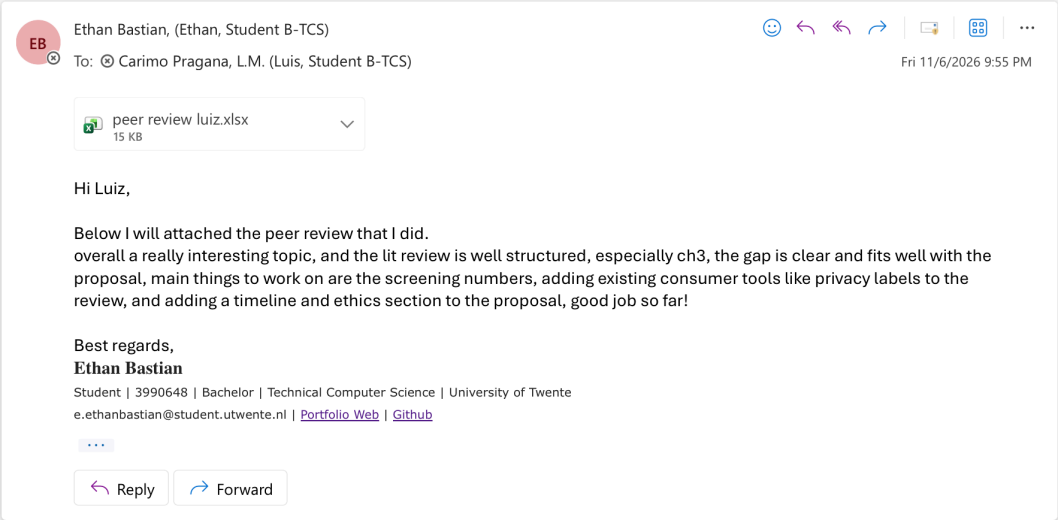


Figure 11: 6 Nov: my peer review sent to Luiz, with a summary of strengths and points to improve.